Yuwei Ren

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EDUCATION

M.S. in Computer Science (B.S./M.S.) – University of California, San Diego | Expected 2026

B.S. in Computer Science – University of California, San Diego | 2023 - 2025 | GPA: 3.8

RESEARCH EXPERIENCE

Early Research Scholars Program (Rose Yu's Group) · UCSD · Sept 2024 – Jun 2025

- Accelerated plasma turbulence simulations by training **TGLF-Neural Network** with **Bayesian Active Learning**, cutting runtime from multi-seconds → sub-second predictions.
- Improved training efficiency **~2.5×** using **Bayesian Active Learning**; trained **370K+ samples** in PyTorch.
- Built **scalable distributed ML pipelines** with Docker + Kubernetes on Nautilus cluster.
- Applied software engineering practices to **physics-guided ML models** in a 3-person research team.

TECHNICAL PROJECTS

LLM Agent for Interactive Game Management | *Python, GPT-4 API, State Machines*

- Built a state-machine LLM agent with **prompt chaining + state tracking** for multi-step decision-making — applied to character creation, dice rolling, and narrative control.
- Demonstrated practical **LLM orchestration**, **API integration**, and deterministic dialogue control — relevant to agentic AI systems.

Autonomous Robot Car – ROS 2 & Qualcomm RB5 Integration | *Python, C++, ROS 2, OpenCV, Linux(Ubuntu 22.04)*

- Built **autonomous navigation** (path planning + dynamic obstacle avoidance) using **SLAM + sensor fusion**
- Integrated **Qualcomm RB5** + mBot Mega → full pipeline of **serial comm**, **motor control**, **visual odometry** in **ROS 2**.

Transitive Closure Computation | *Java, PostgreSQL, Optimized Algorithms*

- Designed and implemented a high-performance back-end system in **Java (OOP)** to compute transitive closure and customer influence paths from large fund transfer data.
- Developed and **optimized** iterative query algorithms in **PostgreSQL** to ensure server-side scalability and efficiency, achieving fast path computation for processing **real-time** distributed data.

Personalized Book Recommendation System | *Hybrid Models, Goodreads*

- Built a hybrid recommender system combining **latent factor models + similarity-based methods**.
- Achieved **MSE 1.42(rating)** and **0.77 (read) accuracy** on the Goodreads dataset.
- Demonstrated ranking, personalization, and model evaluation skills.

Blackjack AI (Reinforcement Learning) | *Monte Carlo, Q-learning*

- Implemented three core Reinforcement Learning (RL) algorithms (**Monte Carlo, Temporal Difference, Q-learning**) to optimize strategic decision-making in a stochastic environment.
- Improved decision-making accuracy by **>27%** compared to random baseline.

PROFESSIONAL SKILLS

- **Machine Learning & AI**: PyTorch, TensorFlow, LLMs / Generative AI (GPT-4), NLP, LLM Agents, CNNs/DNNs, Reinforcement Learning, Bayesian Active Learning, Recommender Systems, Scientific Model Evaluation, SLAM / Robotics AI
- **Systems & Software**: Python (expert), C++, Java, JavaScript, SQL, ROS 2, Docker, Kubernetes, Distributed ML Pipelines, CI/CD, Linux, React, Node.js, Bash/conda scripting, Jest, Puppeteer, Agile workflows